GOODROAD – CITIZEN-POWERED TRAFFIC VIOLATION REPORTING SYSTEM

25-26J-265

Project Proposal Report

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Sri Lanka Institute of Information Technology
Sri Lanka

August 2025

INTELLIGENT TRAFFIC REWARD AND BEHAVIOURAL REINFORCEMENT MANAGEMENT FRAMEWORK

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Declaration page of the candidates & supervisor

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor
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Date: 28th August 2025

Abstract

The Reward Distribution and Behaviour Reinforcement (RDBR) module is a key component of the GoodRoad Citizen-Powered Traffic Violation Reporting System, designed to promote disciplined driving and fair enforcement through dynamic, AI-driven incentives. This module integrates citizen-submitted dashcam footage with public CCTV streams to build comprehensive vehicle behaviour profiles by employing license plate recognition and multi-camera vehicle tracking techniques. A reinforcement learning model optimizes both rewards for citizens who provide valid violation evidence and bonuses for consistently law-abiding drivers, while enabling proportionate sanctions for repeat violators. The system interacts with a mock banking and government service API to simulate lawful restrictions or benefits, ensuring transparency and accountability through a dedicated dashboard. Training is conducted using simulated historical data, focusing on factors such as violation severity, report reliability, and frequency of compliant behaviour. This approach introduces a novel, adaptive incentive structure that not only penalizes misconduct but actively reinforces positive driving practices, contributing to safer and more responsible road usage within the Sri Lankan context.

Keywords: Traffic enforcement, reinforcement learning, reward distribution, behaviour reinforcement, citizen reporting, license plate recognition, multi-camera tracking, mock banking API

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List of Abbreviations

Abbreviation	Description
RDBR	Reward Distribution and Behaviour Reinforcement
AI	Artificial Intelligence
LPR	License Plate Recognition
RL	Reinforcement Learning
API	Application Programming Interface
CCTV	Closed-Circuit Television

1 Introduction

1.1 Overall Introduction

The Reward Distribution and Behaviour Reinforcement (RDBR) module represents a pioneering effort to revolutionize traffic law enforcement in Sri Lanka by combining citizen engagement with advanced AI for dynamic incentive structures. Traditional enforcement mechanisms—typically reliant on manual patrols and fixed infrastructure—often fall short in efficiency, equity, and responsiveness, particularly in densely congested urban settings. Moreover, global implementations of Automated Traffic Enforcement Systems (ATES) have demonstrated improvements in compliance and reductions in crashes, yet they often lack user-driven participation or fail to balance punitive enforcement with positive reinforcement [2], [10].

Vehicle detection and tracking via license plate recognition (LPR)—that is, the automated extraction and interpretation of vehicle registration in real-time—has become central in modern intelligent transportation systems, [5]. Recent developments utilize deep learning approaches to deliver accurate and efficient recognition even under challenging conditions [1], [5], [7]. Equally significant, multi-camera vehicle trajectory tracking enhances enforcement coverage and allows for temporal and spatial profiling of driver behavior [11], [7]. Despite these technological strides, enforcement systems rarely reward compliant behavior or integrate citizen-submitted evidence into a reinforcement-oriented framework.

In parallel, emerging citizen-powered enforcement initiatives such as Israel's Road Guards program have shown that public involvement can substantially amplify enforcement capacity: in just two years, over 3,400 volunteers submitted more than 64,000 traffic violation reports, underscoring the power of technology-enabled public participation in reducing violations [10]. However, these models predominantly focus on punishment and lack adaptive, AI-driven mechanisms to optimize incentives.

The RDBR module therefore proposes a novel, hybrid paradigm: integrating citizen-submitted dashcam evidence and CCTV streams with an AI-driven, reinforcement learning engine to both

penalize repeat offenders and reward consistently compliant drivers. By embedding mock banking and government APIs, the module simulates real-world incentive transactions and sanctions, and presents them transparently through an interactive dashboard. This encourages disciplined driving behaviors while fostering trust and accountability—a dual objective critically absent in many current systems.

1.2 Research Gap

Despite the evolution of traffic enforcement technologies, significant gaps persist. First, many ATES deployments emphasize red-light or speed enforcement but neglect the proactive reinforcement of good behavior, thereby failing to motivate compliance through positive engagement mechanisms [2]. Second, while license plate recognition technology has matured—achieving high accuracy through YOLO-powered detectors and CNN/OCR pipelines [1], [5]—its integration with citizen-sourced reporting remains uncommon. Similarly, high-performance multi-camera tracking systems have proven effective for vehicle re-identification in congested environments [11], [7], but few implementations leverage both data sources synergistically for enforcement.

Additionally, existing citizen-led monitoring programs, while promising, largely remain descriptive and observational in design. The Road Guards program, for instance, validates the efficacy of volunteer-based violation reporting but does not incorporate an adaptive reward system or AI-driven optimization to refine compliance incentives over time [10]. Similar initiatives such as India's Traffic Prahari [12] and Hyderabad's citizen reporting scheme [13] demonstrated that public participation can generate substantial fine revenue and even monetary rewards for reporters, but they primarily rely on fixed or static reward mechanisms and lack adaptive behavior reinforcement features.

Most current systems focus on reactive responses rather than predictive and adaptive strategies that learn from evolving behavioral patterns. There is therefore a pressing gap for a comprehensive framework that integrates advanced detection, multi-view tracking, citizen participation, and reinforcement learning into a unified module that dynamically calibrates rewards and penalties. Particularly in the Sri Lankan context—marked by high urban density,

limited enforcement resources, and growing citizen expectations—such an adaptive, transparent, and scalable system remains unexplored. The RDBR aims to fill this crucial void. A comparative analysis of existing citizen-driven traffic enforcement systems and the proposed framework is presented in Table 1.

Table 1 Comparison with other researches

Research Work	Reward Adaptability	Behaviour Reinforcement	Multi-Source Integration	Sanction Mechanism	Dynamic Incentive Optimization
Road Guards [10]	Fixed	None	No	Manual	No
Traffic Prahari [12]	Fixed cash rewards	None	Limited (manual reports)	Manual fines	Yes
Hyderabad Scheme [13]	Fixed monetary	None	Limited (citizen-only)	Manual fines	No
Proposed RDBR Module	Dynamic (RL-based)	Yes (positive bonuses)	Yes (CCTV + dashcam)	Graduated (RL-driven)	Yes

1.3 Research Problem

The central research problem addressed by the RDBR module is:
How can an AI-driven system integrate citizen participation and automated monitoring to ensure fair, transparent, and adaptive traffic law enforcement that not only penalizes violators but also rewards compliant drivers in real time?
Sub-problems include:

- Identifying authentic violation evidence from diverse video streams (dashcams, CCTV).
- Developing an RL-based model that dynamically adjusts rewards and penalties based on behaviour trends.
- Ensuring data privacy and transparency while interacting with mock banking/government APIs.
- Managing false reports, duplicate reports, or malicious submissions.
- Motivating citizens to participate without incentivizing false claims.

2 Objectives

2.1 Main Objectives

The principal objective of the Reward Distribution and Behaviour Reinforcement (RDBR) module is to create a coherent, evidence-based system that actively shifts traffic-law enforcement from a primarily punitive model to a balanced, adaptive framework that simultaneously discourages dangerous driving and reinforces compliant behaviour. This overarching aim is operationalized by developing an integrated pipeline that accepts citizen-submitted dashcam footage and public CCTV streams, converts them into verifiable evidence using robust computer-vision methods, and then uses an AI-driven decision mechanism to determine proportionate rewards, warnings, or sanctions. Implicit in this objective is the requirement that the system remain transparent, auditable, and resistant to manipulation: every automated action must be traceable to source evidence and accompanied by human-readable justifications suitable for review by authorities and affected drivers.

A secondary but equally important objective is to incentivize constructive citizen participation in traffic safety. The system aims to make it straightforward and rewarding for citizens to report high-quality evidence by offering calibrated rewards that reflect both the severity of the violation and the reliability of the submission. In doing so the RDBR module seeks to increase the spatial and temporal coverage of traffic monitoring without requiring proportional increases in enforcement manpower. This citizen-centric aim also requires building reputation and trust mechanisms that treat repeat high-quality reporters as valuable stakeholders while detecting, down-weighting, and penalizing malicious or low-quality reporting attempts.

From a technical and research perspective, the RDBR module aims to push forward several specific scientific contributions. First, it intends to demonstrate a scalable architecture for fusing multi-source video (dashcam and CCTV) into longitudinal vehicle profiles that support temporal behaviour analysis. Second, it seeks to design and

empirically validate a reinforcement-learning policy that optimizes reward/penalty allocations to maximize long-term compliance rather than short-term reductions in individual infractions. Third, the module aims to establish evaluation protocols and metrics—spanning detection accuracy, evidence verification precision, economic efficiency of rewards, and social outcomes such as measured reductions in repeat offending—that enable objective assessment of impact. Together these objectives position the RDBR as both a practical enforcement tool and a research platform for adaptive public safety interventions.

Finally, the module is designed with operational objectives that translate directly into deployable deliverables: a working prototype that integrates LPR and multi-camera tracking with an evidence verification layer; a trained RL policy and the model-serving infrastructure to apply it at decision time; a user-facing dashboard for citizens, drivers and authorities that provides transparent histories and appeals; and an audit trail system that preserves tamper-evident logs of all decisions. Ethical, legal, and privacy constraints are baked into these objectives—ensuring data minimization, consent handling, secure storage, and a clear appeals process—so that the system's safety and societal benefits are not achieved at the cost of civil liberties or public trust.

2.2 Specific Objectives

- To aggregate citizen-submitted and CCTV footage into a unified vehicle behaviour profile.
- To design a reinforcement learning model that optimizes reward rates for reporters and bonuses for disciplined drivers.
- To simulate lawful sanctions for repeated violations through mock banking/government system integration.
- To create a transparent dashboard for users to view their submissions, rewards, and compliance status.

3 Methodology

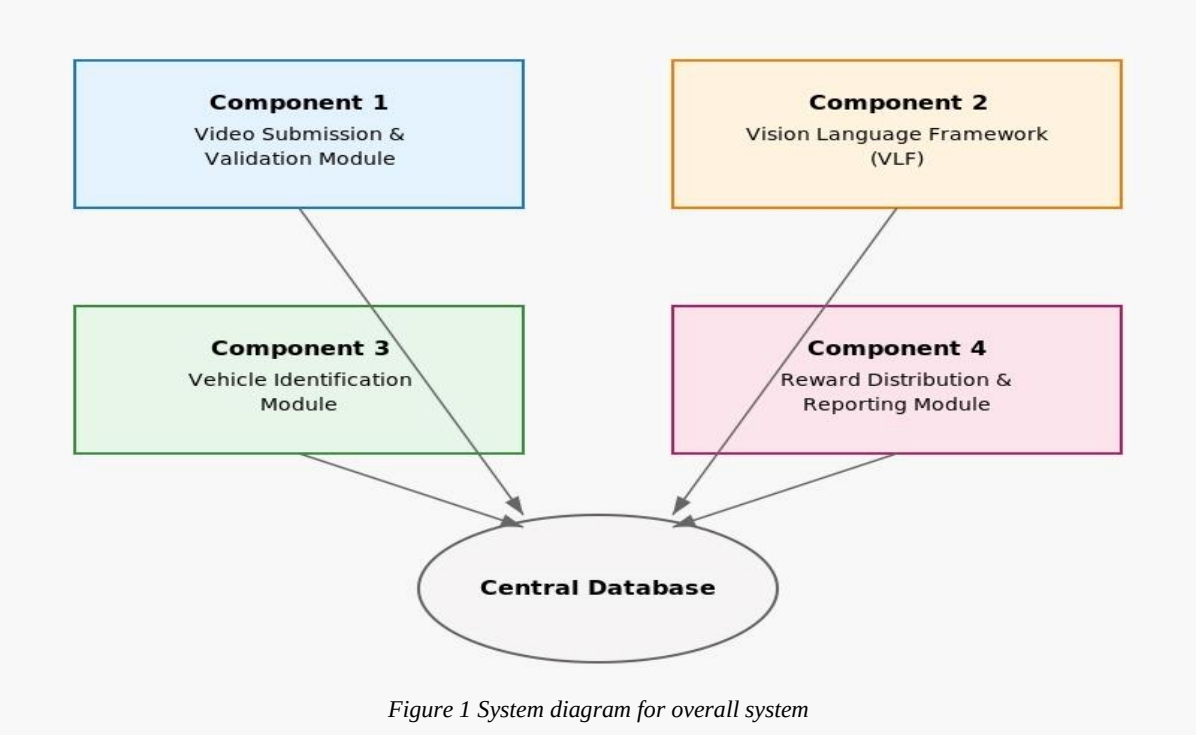

Figure 1 System diagram for overall system

The methodology for the RDBR (Reinforcement-learning-Driven Behavioural Response) module adopts a data-centric, iterative pipeline, combining advanced computer-vision techniques, rigorous evidence validation, and reinforcement-learning (RL)-based policy optimization. At the outset, the process is organized into clearly defined stages: multimodal traffic data acquisition; preprocessing and annotation; LPR (License Plate Recognition) and multi-camera tracking for constructing longitudinal vehicle profiles; evidence verification to assess submission integrity; MDP (Markov Decision Process) formulation of enforcement actions; RL policy training within realistic simulations; and deployment with a human-in-the-loop governance layer.

A key innovation lies in the dynamic incentive optimization through RL. Drawing inspiration from domains like public transportation and urban traffic control, where RL has been successfully applied to optimize dynamic incentives and adaptive control strategies [3], [4], [6], the RDBR module utilizes similar principles to calibrate rewards, warnings, and sanctions dynamically.

For multi-source integration, the system ingests citizen dashcam submissions and live CCTV streams, aligns them via GPS and timestamps, and consolidates them through feature extraction pipelines. Prior studies highlight the benefits of integrating dashcam and CCTV data for real-time vehicle tracking and evidence consolidation [8], supporting the soundness of this approach.

Moreover, to ensure evidence reliability, our design incorporates dashcam forensics best practices—handling timestamp integrity, synchronization, and tamper resilience—to safeguard admissibility and trustworthiness of captured data [3], [9].

Finally, RL-based policy optimization occurs in simulation, mitigating real-world risks. We leverage conservative update techniques (e.g., trust-region or constrained policy optimization), offline evaluation, and staged A/B pilots under human oversight to ensure safe, accountable learning and deployment.

The methodology for the RDBR module is structured into the following key components to provide a clear, systematic overview of the workflow. Each stage corresponds to a critical function in data acquisition, processing, detection, verification, policy optimization, and deployment.

1. Data Acquisition and Consent Management
    - Gather dashcam videos from citizens via secure portals.
    - Ingest public CCTV streams from municipal or partner sources.
    - Collect metadata: timestamps, GPS, device IDs, reporter identity/pseudonym.
    - Include historical violation logs to bootstrap models.
    - Enforce consent, anonymization, and data retention policies.
2. Preprocessing and Annotation
    - Transcode video and extract frames with synchronized timestamps.
    - Align dashcam and CCTV frames via GPS or NTP; map-match to road segments.
    - Apply image enhancement (denoising, stabilization, exposure correction).

- Annotate events using hybrid automated/human pipelines.
- Filter out low-value data (e.g., unreadable plates, non-traffic footage).

3. Vehicle Detection and Multi-Camera Tracking
    - Tier-1: Edge-deployed, low-latency object detection for pre-filtering.
    - Tier-2: High-accuracy, server-side LPR/OCR with deep models tailored to local plates.
    - Build multi-camera tracking with appearance embeddings and spatio-temporal motion modeling.
    - Assign confidence scores to each detection.
4. Evidence Verification
    - Cross-validate OCR outputs across frames and cameras.
    - Confirm geo-temporal consistency (timestamps, GPS vs. road segments).
    - Incorporate auxiliary data if available (e.g., traffic loops, telemetry).
    - Maintain reporter reputation scoring and de-duplicate reports.
    - Flag anomalies for human adjudication.
5. Behavioral Profiling
    - Aggregate verified incidents into vehicle profiles: violation frequency, severity, time patterns, context (weather, road type).
    - Use decay functions to prioritize recent behavior.
    - Feed profiles into RL state and public dashboards.
6. Reinforcement Learning Policy Optimization
    - Define MDP: state = vehicle profile + event history + context + reporter reliability.
    - Actions = reward, compliant acknowledgement, warning, simulated sanction, defer.
    - Design multi-objective reward function: accuracy, social welfare, cost, fairness.
    - Train in simulation with realistic models of driver behaviour and reporting dynamics.
    - Use offline RL and conservative updates, with human oversight and controlled exploration.
    - Validate policies via offline evaluation, pilot A/B tests, and escalation to human review.

7. Decision Service Deployment & Integration
    - Interface with mock banking/government APIs via adapter layers.
    - Record decisions and evidence in tamper-evident audit logs.
    - Provide dashboards with aggregated and case-level views, including appeals workflow.
    - Updates feed into reporter reputations and future training data.
8. Evaluation Metrics
    - Technical: LPR/tracking precision-recall, verification accuracy, latency, duplicate suppression.
    - Policy: cumulative RL reward, violation rate reduction, incentive cost efficiency.
    - Social: reporter engagement, fairness perceptions (via surveys), false report incidence.
    - Employ statistical testing and confidence intervals in pilot rollouts.
9. Deployment, Ethics, and Governance
    - Containerized model serving with edge fallbacks.
    - Secure authentication, encryption, privacy preservation (face/license plate redaction, retention policies).
    - Governance protocols for access, auditing, transparency.
    - Periodic retraining for concept drift; bias audits to ensure equitable impacts.
    - Iterative refinement based on deployment feedback and audits.

Collectively, these stages define a comprehensive, iterative workflow that integrates advanced computer vision, reinforcement learning, and governance mechanisms. The point-wise methodology ensures that each module is designed for accuracy, robustness, and ethical deployment while facilitating monitoring, evaluation, and continuous improvement.

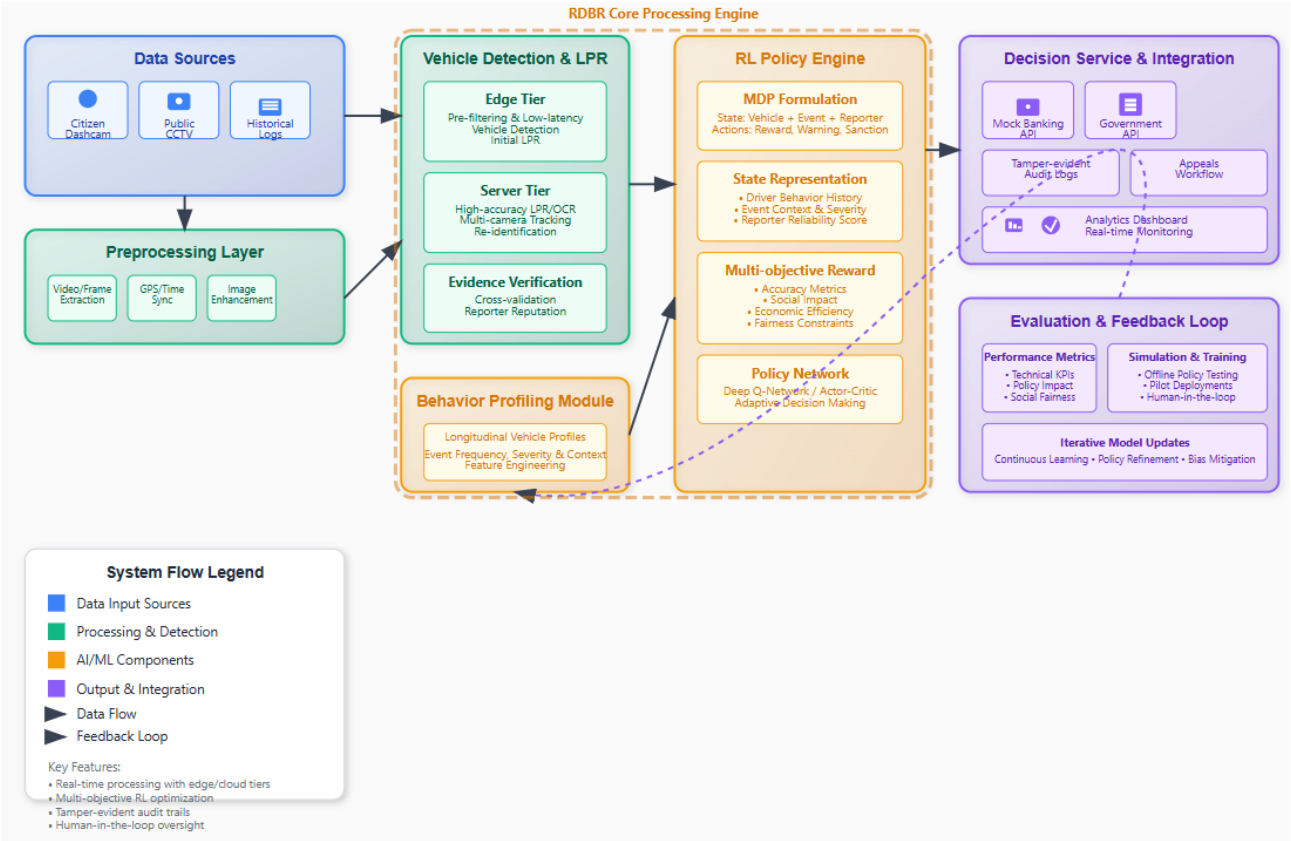

Figure 2 System architecture diagram for the component

4 Project Requirements

4.1 User Requirements

1. Drivers must be able to view compliance points and bonuses.
2. Reporters must track submission and reward status.
3. Authorities must review aggregated data for decisions.

4.2 Software Requirements

4. Web dashboard for reporters and drivers.
5. Backend for reward calculation and behaviour tracking.
6. RL framework (TensorFlow/PyTorch) for policy learning.

4.3 Functional Requirements

1. Citizen Submission Interface
2. License Plate Recognition System
3. Reward Allocation Engine
4. Behaviour Profiling Module
5. Reinforcement Learning Engine
6. Penalty Enforcement System
7. Transparency Dashboard

4.4 Non-Functional Requirements

1. Performance – The system must process up to 1,000 submissions per hour with minimal latency.
2. Scalability – Designed to handle national-level deployment with millions of vehicles.
3. Security – Encrypt citizen submissions using AES-256 encryption, Ensure secure API communication with OAuth 2.0 authentication.
4. Reliability – 99.9% uptime for dashboard services.

4.5 Testing based on the component

This phase ensures that every functionality of the RDBR module operates accurately and without errors. Comprehensive testing will be conducted to validate the module's performance, reliability, and compliance with the specified requirements. Testing techniques include unit testing, integration testing, system testing, and user acceptance testing. Additionally, this phase focuses on identifying potential faults early, resolving them efficiently, and improving the overall quality of the software to ensure a robust and user-friendly solution.

4.5.1 Unit Testing

Unit testing verifies individual components of the RDBR module in isolation. For example, license plate recognition, violation detection, and reward calculation algorithms will be tested independently to ensure they function as intended. This phase helps detect logic errors, incorrect data handling, or inconsistencies before integrating them into the larger system.

4.5.2 Integration Testing

Integration testing examines the interactions between different sub-modules of the RDBR system, such as citizen report validation, reinforcement learning model updates, and mock banking API communication. This phase ensures that data flows seamlessly across modules and that combined functionalities operate without conflicts. Any integration errors, such as incorrect reward distribution or duplicated sanctions, will be identified and resolved during this phase.

4.5.3 System Testing

System testing validates the entire RDBR module as a cohesive system. It ensures that all features—dynamic reward distribution, sanction imposition for repeat violators, citizen dashboard functionality, and driver behaviour tracking—operate together according to the specified requirements. This phase also tests system reliability, performance under high report volumes, and compliance with Sri Lankan traffic enforcement regulations.

4.5.4 User Acceptance Testing

In this final testing phase, end-users—including government authorities, law enforcement officers, and selected citizen reporters—will evaluate the RDBR system against its original objectives. The purpose is to verify usability, transparency of the reward and penalty mechanisms, and overall practicality in real-world scenarios. UAT feedback will guide any final adjustments before full deployment.

5 Gantt Chart

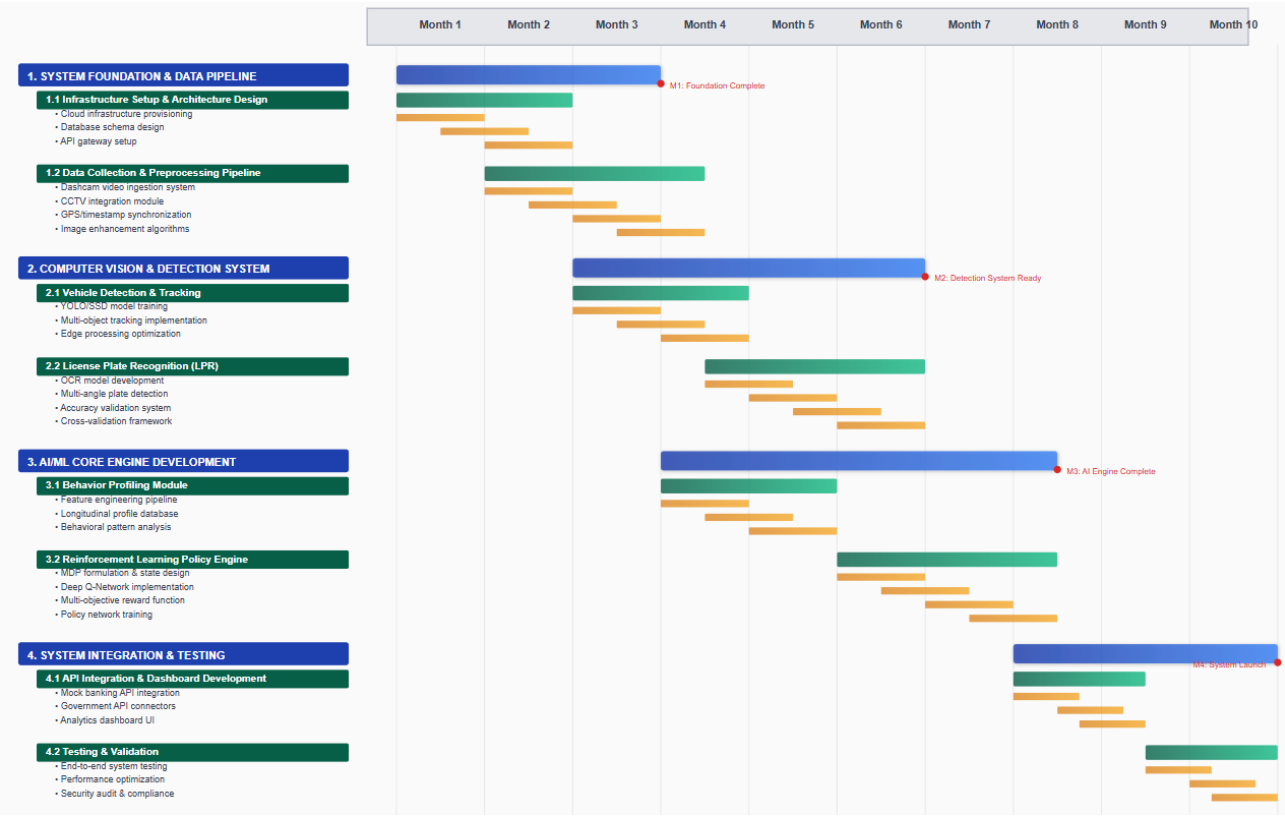

Figure 3 Gantt Chart

6 Work breakdown Chart

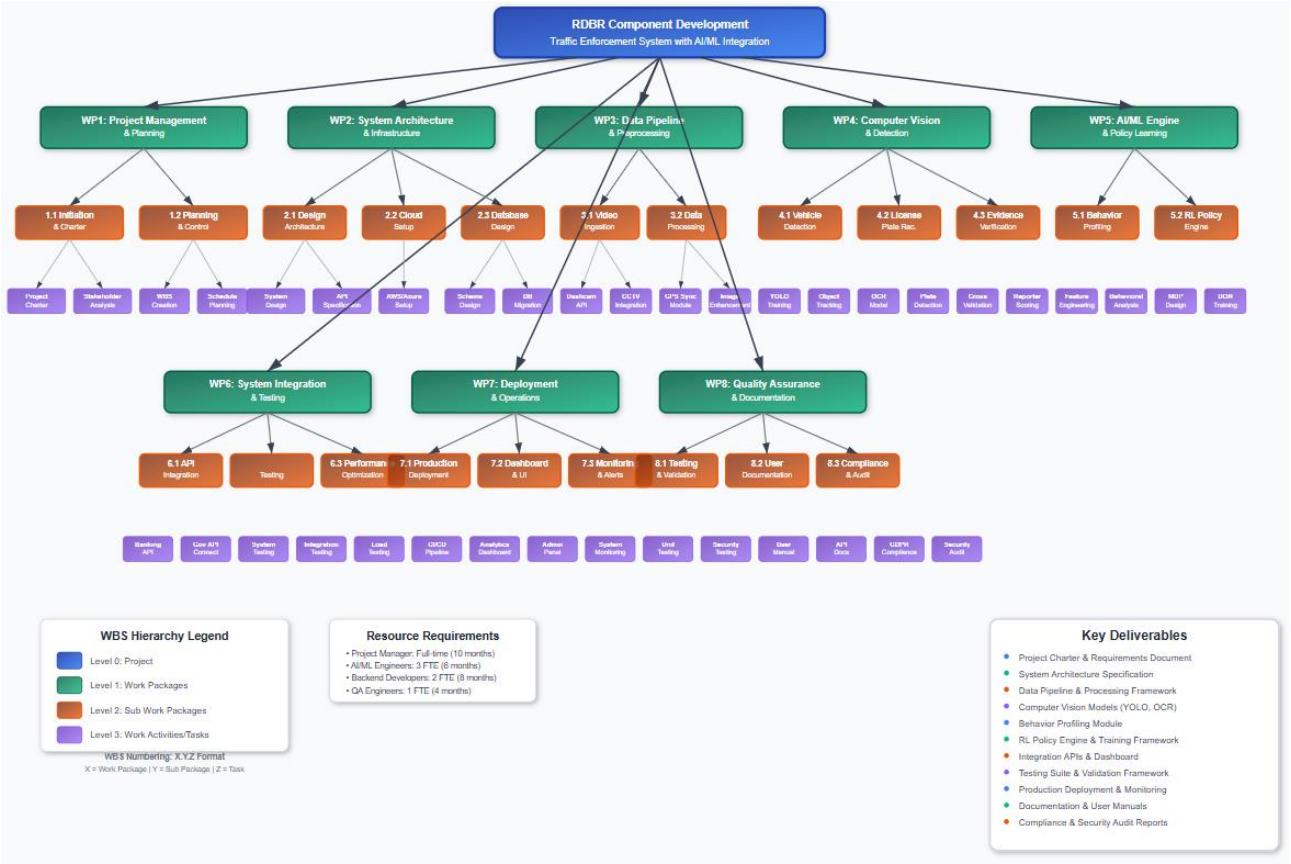

Figure 4 Work breakdown structure

7 Budget and Commercialization

Table 2 Budget

Component	Cost
Datasets (CCTV + Dashcam + Historical Logs)	Rs. 15,000
Travelling / Data Collection Costs	Rs. 10,000
Deployment & Integration (API adapters, dashboards, pilot servers)	Cost Per Usage

7.1 Commercialization Strategy

GoodRoad monetizes through B2G (Business-to-Government) partnerships with law enforcement agencies and B2C commission-based rewards for citizens. Revenue streams include:

- Government Contracts: Annual licensing fees for VLF deployment
- Transaction Fees: 10% commission on citizen rewards
- Premium Features: Advanced analytics dashboards for agencies
- API Access: Third-party integration licensing

7.2 Target Audience

- Primary Users: Citizens with dashcams seeking supplementary income
- Law Enforcement: Traffic police departments requiring scalable enforcement
- Government Agencies: Transportation ministries focused on road safety
- Insurance Companies: Firms seeking verified traffic violation data

7.3 Market Space

- Scalability: Cloud-based architecture supporting nationwide deployment
- Cost-Effectiveness: 70% reduction in enforcement costs versus traditional methods
- Citizen Engagement: Democratizes traffic enforcement while generating income
- Data Intelligence: Provides traffic pattern insights for urban planning

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